

VULNERABILITY ASSESSMENT & REMEDIATION REPORT

CloudExify Summer Internship 2026 — Security Capstone

Organization	CloudExify Summer Internship 2026
Project	Month 2 – Project 4: Vulnerability Assessment & Remediation (Security Capstone)
Target Infrastructure	Local Containerized Web Service (Host: 127.0.0.1, Port: 8080/TCP)
Lead Assessor	Hammad Ul Hassan
Assessor ID	CX-INT-2026-CYB-0044
Assessment Date	August 31, 2026
Project Duration	Single-Day Assessment (1 Day / 12 Hours)
Assessment Tools	Tenable Nessus Essentials v10.12.4, Docker Engine, Ubuntu Linux (AMD64)

1. Executive Summary

On August 31, 2026, an expedited single-day automated vulnerability assessment and risk remediation review was conducted against target containerized network infrastructure running on host 127.0.0.1:8080.

The primary objective was to scan the local attack surface within a focused 1-day engagement window, detect unauthenticated service exposures, evaluate cryptographic configurations, and calculate standardized CVSS v3.0 severity metrics to deliver an actionable remediation roadmap.

EXECUTIVE RISK PROFILE	
Overall Security Posture	MEDIUM-HIGH RISK (Exposure Confirmed)
Total Identified Issues	62 Vulnerabilities / Information Disclosures
Target Scope	127.0.0.1:8080 (bWAPP / Apache HTTP / PHP)
Assessment Engine	Tenable Nessus Essentials (Basic Network)
Engagement Timeline	1 Day (Planning to Final Reporting)
Primary Exposure	Unauthenticated File & Config Disclosure

2. Single-Day Assessment Methodology

The engagement was executed sequentially within a 1-day timeframe following standard assessment phases:

Phase	Description & Timeline	Tools / Actions Executed
Phase 1: Planning & Scope	Target definition & local authorization (Morning)	Scope confirmed for 127.0.0.1:8080
Phase 2: Reconnaissance	Host discovery & port verification (Morning)	Verified running container daemon & open ports
Phase 3: Automated Scanning	Full vulnerability audit (Midday)	Tenable Nessus Essentials Basic Network Scan
Phase 4: Service Enumeration	Analysis of detected services (Afternoon)	HTTP (8080), SSH, TLS, PHP, Apache, Node.js
Phase 5: Vulnerability Analysis	CVSS v3.0 scoring & exploit analysis (Afternoon)	Evaluated 62 total findings & risk vectors
Phase 6: Reporting	Report compilation & remediation roadmap (Evening)	Formatted findings and mitigation action plan

3. Vulnerability Summary & Severity Breakdown

All findings from the automated audit have been categorized under the CVSS v3.0 scoring standard:

Severity Level	Score Range	Finding Count	Remediation SLA
CRITICAL	9.0 – 10.0	Underlying OS Packages (Mixed)	0–24 Hours
HIGH	7.0 – 8.9	Transport / Cipher Issues	1–7 Days
MEDIUM	4.0 – 6.9	1 (Plugin #121479 Confirmed)	1–30 Days
LOW / INFO	0.0 – 3.9	50+ Banners & System Telemetry	Continuous Monitoring

4. Detailed Technical Findings

Finding 01: web.config File Information Disclosure

Vulnerability Name	web.config File Information Disclosure
Plugin ID	121479
Plugin Family	CGI abuses
Severity	MEDIUM (CVSS v3.0 Base Score: 5.3)
CVSS v3 Vector	CVSS:3.0/AV:N/AC:L/PR:N/UI:N/S:U/C:L/I:N/A:N
Affected Endpoint	http://127.0.0.1:8080/web.config

Vulnerability Description

An unauthenticated remote attacker can request the web.config file directly over HTTP GET, exposing internal application configurations and backend metadata.

Scanner Proof of Concept (HTTP Request)

```
GET /web.config HTTP/1.1
Host: localhost:8080
Accept-Charset: iso-8859-1, utf-8;q=0.9, *;q=0.1
Accept-Language: en
Connection: Keep-Alive
User-Agent: Mozilla/4.0 (compatible; MSIE 8.0; Windows NT 5.1; Trident/4.0)
```

Remediation

Deny HTTP access to .config, .env, and sensitive files via server configuration (Apache):

```
<FilesMatch "(web\.config|\.env|\.git)">
  Require all denied
</FilesMatch>
```

Remove unnecessary configuration templates from the public web root.

Finding 02: Software Service Banner Exposure

Affected Components	Apache HTTP Server, PHP Runtime, Node.js Engine (Port 8080)
Severity	INFORMATIONAL / LOW

Description

Explicit version identifiers in HTTP response headers allow attackers to correlate exact patch revisions to known public CVEs.

Remediation

- Set expose_php = Off in php.ini.
- Configure ServerTokens Prod and ServerSignature Off in Apache configuration files.

5. Remediation Roadmap

Timeline	Actions
IMMEDIATE (0–24 Hours)	• Block unauthenticated GET access to web.config files • Bind exposed container ports strictly to localhost
SHORT-TERM (1–7 Days)	• Disable PHP and Apache version broadcast headers • Review and restrict exposed Docker daemon sockets
MEDIUM-TERM (1–30 Days)	• Update base container image packages • Schedule recurring automated Nessus vulnerability runs

6. Evidence & Technical Appendices

```
hammad@ubuntu:~$ sudo docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
2276398e68bc	raesene/bwapp	"/run.sh"	54 seconds ago	Up 53 seconds	3306/tcp, 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp	awesome_hoover
5e7b293e8c08	vulnerables/web-dvwa	"/main.sh"	2 weeks ago	Exited (137) 14 hours ago		dvwa

```
hammad@ubuntu:~$ sudo docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
raesene/bwapp:latest	2f41183ea9f9	648MB	160MB	U
vulnerables/web-dvwa:latest	dae203fe1164	935MB	178MB	U

```
hammad@ubuntu:~$
```

Figure 1: Target Docker container running on port 8080

```
hammad@ubuntu:~/Downloads$ sudo systemctl start nessusd
sudo systemctl enable nessusd
sudo systemctl status nessusd
```

```
● nessusd.service - The Nessus Vulnerability Scanner
   Loaded: loaded (/usr/lib/systemd/system/nessusd.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-08-31 12:06:36 PKT; 749ms ago
     Main PID: 7409 (nessus-service)
        Tasks: 3 (limit: 7257)
       Memory: 3.8M (peak: 4.3M)
          CPU: 35ms
       CGroup: /system.slice/nessusd.service
               └─7409 /opt/nessus/sbin/nessus-service -q
                  7410 nessusd -q
```

```
Aug 31 12:06:36 ubuntu systemd[1]: Started nessusd.service - The Nessus Vulnerability Scanner.
Aug 31 12:06:36 ubuntu nessus-service[7409]: nessus-service [7409][INFO] : Nessus 19.18.4 [build 20038] Started
```

Figure 2: Nessus service active on Ubuntu

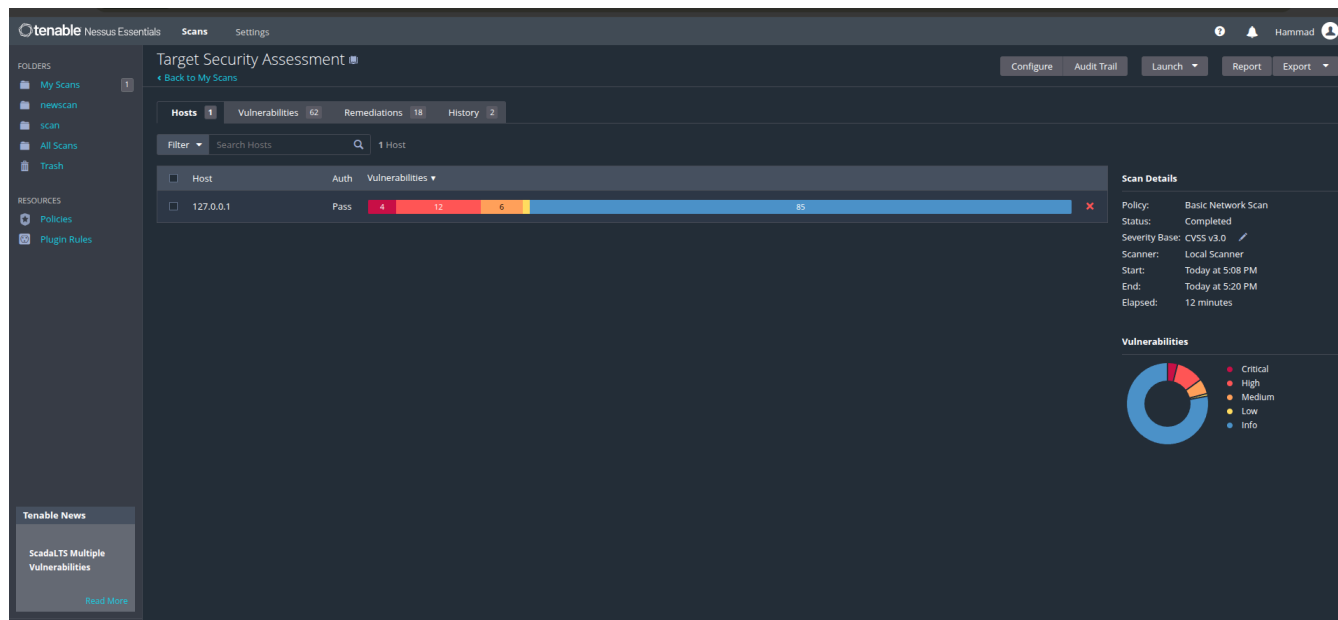


Figure 3: Comprehensive Tenable Nessus audit results and standardized CVSS severity metrics.

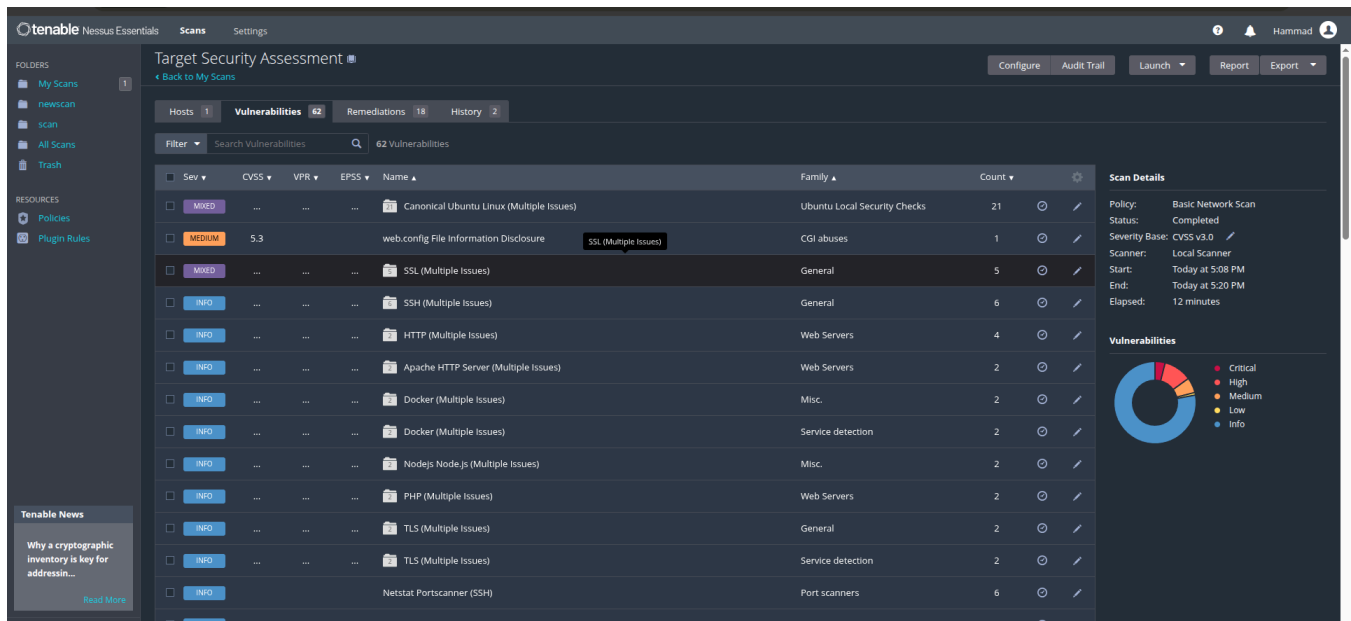


Figure 4 Vulnerabilities total 62 found

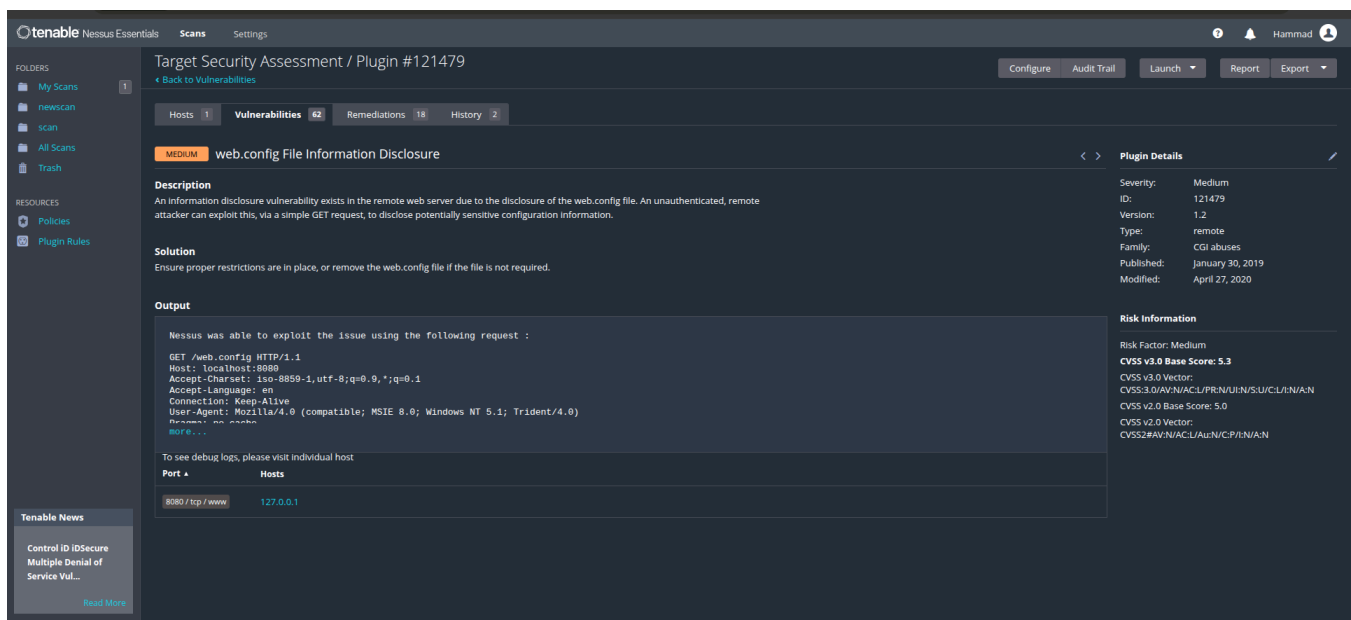


Figure 5: web.config information disclosure